SINCHANA RAJ G

SDE | Data Analyst | Applied AI/ML

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PROFILE SUMMARY

Full-stack engineer who ships complete products, not prototypes — frontend to database to model. At BEL, built and deployed a three-service clinical risk-prediction platform (React, Spring Boot, FastAPI/XGBoost) with an explainability layer that turns model output into plain-language decisions for non-technical stakeholders. Outside of work I build the same way end-to-end: a GenAI mock-interview product, an explainable recommendation engine, a churn-prediction pipeline that ends in a business playbook, and a product teardown backed by real analysis of 1,000+ user reviews. Comfortable owning a feature from schema design to UI to explaining the "why" behind a model's call.

INTERNSHIP EXPERIENCE

Data & ML Engineering Intern | *Bharat Electronics Limited (BEL) — Defence PSU, Govt. of India*Jan – Apr 2026

- **Built CardioPredict, a three-service clinical risk-prediction platform:** React frontend, Spring Boot REST API with JWT role-based auth, and a FastAPI microservice serving an XGBoost model — architecture mirrors a production multi-service system, not a notebook.
- **Trained an XGBoost classifier reaching 89.4% test AUC-ROC** on clinical data, then added a SHAP explainability layer so every prediction ships with ranked risk drivers — translating model output into plain-language narratives for non-technical stakeholders.
- **Designed the PostgreSQL schema and REST contracts** across auth, patient-records, and prediction services; containerized the full stack with Docker Compose for reproducible deployment and documented the APIs for cross-team handover.
- **Shipped three role-specific views** (patient self-service, admin management, public researcher analytics) on top of one shared prediction pipeline — same model, three different audiences.

PROJECTS

Vance — AI Mock Interviewer — *React, TypeScript, FastAPI, PostgreSQL, OpenAI API*

- **Built a full-stack GenAI product** — paste a job description, get an AI-generated tailored mock interview with live streamed feedback and a per-role Interview Readiness Score tracked across sessions.
- **Designed a 3-stage LLM pipeline:** JD analysis (skills/seniority extraction) → adaptive question generation (technical, behavioral, system-design) → structured evaluation (accuracy, communication, gaps, model answer) parsed into a normalized Postgres schema.
- **Streamed AI feedback via Server-Sent Events** on a FastAPI + SQLAlchemy + Alembic backend, so candidates see feedback progressively instead of waiting on a full model response.

Jenny — Explainable Movie Recommendation Engine — *Python, scikit-learn, SHAP, FastAPI*

- **Built a collaborative-filtering recommender** (matrix factorization) on the real MovieLens-100K dataset (943 users, 1,682 items, 100K ratings), reaching Precision@10 of 0.168.
- **Solved the "why was this recommended" problem:** trained a surrogate gradient-boosted model on interpretable features (genre overlap, recency, popularity) to mimic the CF model's latent-factor scores at 0.93 fidelity (R^2), then applied SHAP for a plain-English explanation per user-item pair.
- **Exposed recommendations and explanations via a FastAPI service** — an explainability pattern directly reusable for recommendation trust/UX in streaming, e-commerce, or edtech products.

SaaS Churn Prediction + Churn Playbook — *Python, scikit-learn, SQL, Excel*

- **Built an end-to-end churn pipeline** on ~4,000 SaaS customers across 29 monthly snapshots (~45K customer-months): trend feature engineering (usage decay, login gaps, support incidents) → time-based-split classifier (no future-data leakage) → revenue-weighted risk tiers.
- **Reached 0.72 ROC-AUC on held-out future months** and identified consecutive-month usage decline as the single strongest leading indicator of churn, well ahead of point-in-time usage or ticket volume.
- **Delivered a 5-tier Revenue-at-Risk system** (flagged $192K of MRR in the Critical tier) plus a Churn Playbook mapping each tier to a specific trigger and action — e.g. inactive 7+ days → auto reminder; high-value + high-risk → escalate to CS — and an Excel dashboard with live formulas and native charts.

Duolingo PM Teardown — Solving the Engagement ≠ Conversion Gap — *Python, NLTK, scikit-learn*

- **Ran a product case study on Duolingo's free-to-paid conversion problem:** scraped and sentiment-scored 1,040 App Store reviews (VADER), then clustered negative reviews with TF-IDF + KMeans to surface real complaint themes at scale instead of reading reviews one by one.
- **Found the largest complaint cluster (43% of reviews analyzed)** centered on the hearts/energy system and subscription friction — turned this into evidence-backed product bets instead of guesses.
- **Wrote a 4-page PRD** translating findings into three prioritized product bets (contextual upgrade nudges, tiered streak insurance, social features behind a soft paywall), each with a hypothesis, RICE-style prioritization, and success/guardrail metrics.

TECHNICAL SKILLS

- **Languages:** Python, SQL (PostgreSQL), Java, TypeScript/JavaScript, R
- **Full-Stack:** React, FastAPI, Spring Boot, REST APIs, JWT Auth, SQLAlchemy, Alembic
- **ML & AI:** XGBoost, scikit-learn, SHAP, Collaborative Filtering/SVD, NLP (VADER, TF-IDF), OpenAI API integration, A/B Testing concepts
- **Data & BI:** Pandas, NumPy, Power BI, Tableau, Excel, Cohort & Churn Analysis
- **Engineering & Product:** Docker, Docker Compose, Git, PRD writing, RICE prioritization, stakeholder-ready reporting

EDUCATION

B.E. (Computer Science & Engineering) Malnad College of Engineering 2022 – 2026
Pre-University (PCMB) Gnana Jyothi PU College 2021 – 2022

CERTIFICATIONS

- Business Intelligence & Analytics — NPTEL (2025)
- Human Computer Interaction — NPTEL (2025)
- R Programming — Infosys Springboard (2025)
- Introduction to IoT & Industry 4.0 — NPTEL (2025)